Sets, Functions, etc..

By a set we mean an unordered collection of objects. We use various notations:

$$A = \{1, 3, 5, 7, 8\},$$

$$B = \{n \in \mathbb{N} : n \text{ is odd}\}$$

$$A = \{(x, x^2 - 2) : x \in \mathbb{R}\}$$

Definition
$A \subseteq B$ if $\forall x(x \in A) \rightarrow (x \in B)$ is true. In other words, $A \subseteq B$ if $x \in B$ whenever $x \in A$.

Definition
$A = B$ if $A \subseteq B$ and $B \subseteq A$. Therefore, $A = B$ iff $\forall x(x \in A) \longleftrightarrow (x \in B)$ is true.

Let X be a set. If $A, B \subseteq X$, we define

1. $A \cap B = \{x \in X : x \in A \text{ and } x \in B\}$
2. $A \cup B = \{x \in X : x \in A \text{ or } x \in B\}$
3. $A - B = \{x \in A : x \notin B\}$
4. $\overline{A} = A^c = \{x \in X : x \notin A\} = X - A$.

Therefore,

1. $x \in A \cup B$ iff $(x \in A) \vee (x \in B)$ is true.
2. $x \in A \cap B$ iff $(x \in A) \wedge (x \in B)$ is true.
3. $x \in A - B$ iff $(x \in A) \wedge (x \notin B)$ is true.
4. $x \in \overline{A}$ iff $x \notin A$.

Wenn diagrams are very useful!!

Cartesian Products, Relations

Let A, B be sets. Then $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$. A relation from A to B is subset of $A \times B$.

Example

$C = \{(x, y) : x, y \in \mathbb{R} : x^2 + y^2 = 1\}$ is a relation from $\mathbb{R}$ to $\mathbb{R}$.

If R is a relation from A to B, and if $(x, y) \in R$, we say that x is an input of R and y is an output of R corresponding to x. If R is a relation from A to B, the domain of R is the set of all inputs of R. The range of R is the set of all outputs of R.
A function is a special type of a relation:

Definition

A function $f : A \to B$ is a relation such that the domain of f is the whole of A and such that every input of f has a unique output.

Example

1. The relation $\{(x, 2x) : x \in \mathbb{R}\}$ is a function with domain $\mathbb{R}$. Such a function is represented by $f(x) = 2x$.
2. $\{(x, y) : x^2 + y^2 = 1\}$ is not a function.
3. $\{x; \sqrt{1 - x^2}) : x \in [-1, 1]\}$ is a function. $(g(x) = \sqrt{1 - x^2})$

Definition

A function $f : A \to B$ is said to be 1-1 if different outputs corrsepond to different inputs. In other words, f is 1-1 iff $\forall x \forall y (f(x) = f(y)) \to (x = y)$ is true. $f : A \to B$ is said to be onto if every $y \in B$ is an input. In other words, f is onto iff $\forall y \exists x : y = f(x)$.

Example

$f(x) = x^2$ is not 1-1, whereas $g(x) = x^3$ is 1-1. x^2 is not onto, whereas x^3 is onto. (Both f and g are considered as functions : $\mathbb{R} \to \mathbb{R}$)

Definition

Let $f : A \to B$ and $g : B \to C$ be functions. The composition of f and g is the function $h(x) = g(f(x))$ for all $x \in A$. Thus, $h = g \circ f$ is a function : $A \to C$.

Definition

Suppose that $f : A \to B$ is a function. If $g : B \to A$ is a function with $f(g(y)) = y$ for all $y \in B$ and $g(f(x)) = x$ for all $x \in A$, we say that g is the inverse function of f and is denoted by f^{-1}.

Lemma

1. *If $g \circ f$ is onto, then g is onto.*
2. *If $g \circ f$ is 1-1, then f is 1-1.*

Proof.

□

Therefore, if f has an inverse, then it must both 1-1 and onto. The converse is also true.

Theorem

$f : A \to B$ has an inverse iff it is 1-1 and onto.

If $f : A \to B$ is 1-1,and onto, we sometimes say that f is a 1-1 correspondence between A and B.

Exercise

If $f : A \to B$ is onto, then there exists a $g : B \to A$ such that $F(g(y)) = y$ for all $y \in B$. If $f : A \to B$ is 1-1, then there is a function g from B onto A such that $g(f(x)) = x$ for all $x \in A$.

Cardinality of a set

For $n \geq 1$, let $[n] = \{1, 2, ..., n\}$.

Definition

We say that the cardinality of a set S is n if there is a 1-1 correspondence between S and $[n]$. In this case we also say that S has n elements. If a set S has n elements for some $n \in \mathbb{N}$, we say that S is finite. If a set is not finite, we say that S is an infinite set. If there is a 1-1 correspondence between two sets S and T, we say that S and T have the same cardinality and write $|S| = |T|$. If S is finite, let $|S|$ denote the size, or cardinality of S.

If there is a 1-1 function from A to B, we write $|A| \leq |B|$.

Theorem (Schröder-Bernstein)

If $|A| \leq |B|$ *an* $|B| \leq |A|$, *we have* $|A| = |B|$.

The proof is omitted.

Fact

1. $\mathbb{Z}$ *and* $\mathbb{N}$ *have the same cardinality.*
2. $\mathbb{Q}$ *and* $\mathbb{N}$ *have the same cardinality.*

If a set S is infinte and has the same cardinality with $\mathbb{N}$, we say that S is countable, or countably infinite. Therefore, $\mathbb{Z}, \mathbb{Q}$ are countable. The following proposition shows that $\mathcal{P}(\mathbb{N})$ is not countable. Therefore there are infinite sets which are not countable.

Proposition

Let S be a set. There is no 1-1 correspondence between $\mathcal{P}(S)$ and S.

This shows that $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$. Similarly $|\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))|$ and so on.. There are infinitely many infinite cardinalities.

Proof.

Suppose that $\varphi : S \to \mathcal{P}(S)$ is a 1-1 correspondence. Let $A = \{x \in S : x \notin \varphi(x)\}$. Then, $A \subseteq S$. Therefore, $A \in \mathcal{P}(S)$. Since φ is a 1-1 correspondence, there exists a unique $a \in S$ such that $\varphi(a) = A$. Now, we see that

$$a \in A \implies a \in \varphi(A) \implies a \notin A$$

and, similarly,

$$a \notin A \implies a \notin \varphi(a) \implies a \in A.$$

This contradiction proves the claim. □

Proposition

Let X be the set of functions $f : \mathbb{N} \to \{0, 1\}$. Then X is uncountable.

Proof.

Suppose that X is countable. Then we can write X as a list

$$\{f_1, f_2, f_3, ..., f_n, ...\} = \{f_n : n \in \mathbb{N}\}.$$

Define $g(n) = 1 - f_n(n)$ for each n. This g belongs to X but it is not on the list. Therefore we can not list all elements of X. This proves that X is not countable. □

Conjecture (Continuum Hypothesis)

There is no cardinality between $|\mathbb{N}|$ and $|\mathcal{P}(\mathbb{N})|$.

Sequences

By a sequence in a set X, we mean a function $f : \mathbb{N} \to X$. We write (x_n) for a sequence f, where $x_n = f(n)$ for each $n \in \mathbb{N}$. If $f : [n] \to X$ is a function, we say that we have a finite sequence. Let f be a finite or infinite sequence.

$$\sum_{i=0}^{n} f(i) = f(0) + f(1) + f(2) + ... + f(n).$$

Similarly we define

$$\prod_{i=0}^{n} f(i) = f(0).f(1).f(2)....f(n).$$

For example, $n! = \prod_{i=1}^{n} i$.
0! is defined to be 1.

Mathematical Induction

Suppose that $\mathbb{N} = \{1, 2, 3, \cdots\}$. Let $A \subseteq \mathbb{N}$.
Principle of Mathematical Induction: If $1 \in A$ and $n + 1 \in A$ whenever $n \in A$, then $A = \mathbb{N}$.

Example

For $n \in \mathbb{N}$ we have

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof.

Let A be the set of all $n \in \mathbb{N}$ such that the equation above is true. We want to show that $A = \mathbb{N}$.

1. $n = 1$: This is trivial.
2. Suppose that $\sum_{i=1}^{n} i = n(n+1)/2$. Then we have

$$\sum_{i=1}^{n+1} i = \left(\sum_{i=1}^{n} i \right) + (n+1) = (n(n+1)/2) + (n+1) = (n+1)(n+2)/2.$$

The proof is complete.

□

Example

If $n \geq 1$, we have

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1.$$

Proof.

1. $n = 1$: This case is trivial.
2. Suppose that the equality above is true for $n \in \mathbb{N}$. Then we have

$$\sum_{i=0}^{n+1} 2^i = \left(\sum_{i=0}^{n} 2^i \right) + 2^{n+1} = 2^{n+2} - 1.$$

□

Example

If $r \neq 1$ and $n \in \mathbb{N}$, we have

$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1}.$$

Proof.

1. The case $n = 1$ is trivial.
2. Suppose we have the equality for n. Then, we have

$$1 + r + \cdots + r^{n+1} = \frac{r^{n+1} - 1}{r - 1} + r^{n+1} = \frac{r^{n+2} - 1}{r - 1}.$$

□

Example

1. For $n \geq 1$ we have $n < 2^n$.
2. For $n \geq 4$ we have $2^n < n!$
3. For $j \geq 1$ put

$$H_j = 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{j}.$$

   We claim that

$$H_{2^n} \geq 1 + \frac{n}{2}$$

   whenever $n \geq 1$.
4. For $n \geq 1$ we claim that $n^3 - n$ is divisible by 3.

Example

Show that $7^{n+2} + 8^{2n+1}$ is always divisible by 57 whenever n is a nonnegative integer.

Proof.

1. $n = 0$ is trivial.
2. Suppose that $7^{n+2} + 8^{2n+1} = 57K$ for some integer K. Then,

$$7^{n+3} + 8^{2(n+1)+1} = 7.7^{n+2} + 64.8^{2n+1} = 57(7K + 8^{2n+1}).$$

□

Example

If a set S has n elements, then its power set $\mathcal{P}(S)$ has 2^n elements.

Proof.

1. $P(0)$ is true.
2. Suppose that $P(n)$ is true. Let T be a set with $n+1$ elements, let $x \in T$ and $S = T \backslash \{x\}$. Then, we have $\mathcal{P}(T) = \mathcal{P}(S) + \{A \cup \{x\} : A \in \mathcal{P}(S)\}$. Therefore $\mathcal{P}(T)$ has $2.2^n = 2^{n+1}$ elements.

□

Fact (Strong Induction)

Suppose that

1. $P(0)$ *is true.*
2. *For every positive* k *the conditional statement*

$$P(1) \wedge P(2) \wedge ... \wedge P(k) \rightarrow P(k+1)$$

is true.

Then, the statement $P(n)$ *is true for every nonnegative integer.*

Example

If n is an integer greater than 1, then n can be written as a product of primes.

Proof.

1. $P(2)$ is true.
2. Suppose that $P(2), ...P(n)$ are all true. If $n+1$ is prime, there is nothing to prove! If $n+1$ is not a prime, then $n+1 = ab$ where a and b are integers greater than 1 and smaller than $n+1$. By hypthesis both a and b are products of primes. The result follows.

□

Example

Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.

The proof will be by strong induction on n. The case $n = 12 = 3 \cdot 4$ is easy. The cases $13 = 2 \cdot 4 + 1 \cdot 5$, $14 = 1 \cdot 4 + 2 \cdot 5$, $15 = 3 \cdot 5$, $16 = 4 \cdot 4$ are also easy. Suppose we know that $n \geq 16$ and the statement is true for $k = 16, 17, ..., n$. Then the result is true for $n - 4$. This means that there are natural numbers x and y such that $n - 4 = x \cdot 4 + y \cdot 5$. But, then, it follows that $n = (x + 1) \cdot 4 + y \cdot 5$. This completes the proof.

Well-ordering Principle

Fact

If $A \subseteq \mathbb{N}$ is non-empty, then A has a smallest element.

Example

If a is an integer and d is a positive integer, then there are unique integers q and r with $0 \leq r < d$ and $a = dq + r$.

Proof.

Let $A = \{a - dq : q \in \mathbb{Z}\} \cap \mathbb{N}$. A is non-empty because $a + q$ or $a - q$ is an elment of A. By the Well-Ordering Principle, A has a least element, say r. Then $r \geq 0$ and $a = dq + r$ for some $q \in \mathbb{Z}$. This r should satisfy $r < d$: Otherwise $r - d$ would be in A. Uniqueness is also easy. □

Basic Counting

Product Rule Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure.

Example

A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Example

There are 32 computers in a data center in the cloud. Each of these computers has 24 ports. How many different computer ports are there in this data center?

Example

How many different bit strings of length seven are there?

Example

How many functions are there from a set with m elements to a set with n elements?

Example

How many 1-1 functions are there from a set with m elements to a set with n elements?

Solution:

The answer is $n \cdot (n-1) \cdot (n-1) \cdots (n-m+1) = \frac{n!}{(n-m)!}$. We remark that if $n < m$ some of these terms will be 0, hence there will be no 1-1 function from A to B. $\square$

Example

Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$.

Solution:

□

General form:

$$|A_1 \times A_2 \times \cdots A_m| = |A_1| \cdot |A_2| \cdots |A_m|.$$

If a task can be done either in one of n_1 ways or in one of n_2 ways, where none of the set of n_1 ways is the same as any of the set of n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Example

Suppose that either a member of the mathematics faculty or a student who is a mathematics ma- jor is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?

Example

A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

General Form:

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = |A_1| + |A_2| + \cdots + |A_n|$$

whenever $A_i \cap A_j = \emptyset$ for $i \neq j$.

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Example

How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution:

Let A be the set of all strings of length 8 which starts with 1, B be the set of strings of length 8 which ends in 00. Then, we have $|A| = 2^7$, $|B| = 2^6$ and $|A \cap B| = 2^5$. Therefore, we have $|A \cup B| = 2^7 + 2^6 - 2^5 = 128 + 64 - 32 = 160$. □

Example

A computer company receives 350 applications from college graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

Solution:

Let C be the set of applicants with major in computer science, B be the set of applicants with major in business. Then we have $|C| = 220$. $|B| = 147$ and $|B \cap C| = 51$. Therefore, we have $|B \cup C| = 220 + 147 - 51 = 316$. We also have $|U| = 350$. Therefore, $|(B \cup C)^c| = 350 - 316 = 34$. □

There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

Example

Suppose that an automated system has been developed that counts the legs of cows in a pasture. Suppose that this system has determined that in a farmer's pasture there are exactly 572 legs. How many cows are there is this pasture, assuming that each cow has four legs and that there are no other animals present?

Example

How many different ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left neighbor and the same right neighbor?

Example

How many bit strings of length four do not have two consecutive 1s?

Example

A playoff between two teams consists of at most five games. The first team that wins three games wins the playoff. In how many different ways can the playoff occur?

Example

How many positive integers not exceeding 100 are divisible either by 4 or by 6?

Example

How many bit strings of length 10 either begin with three 0s or end with two 0s?

Example

How many functions are there from the set $\{1, 2, \ldots, n\}$, where n is a positive integer, to the set $\{0, 1\}$

1. that are one-to-one?
2. that assign 0 to both 1 and n?
3. that assign 1 to exactly one of the positive integers less than n?

Example

Show that a natural number $n \geq 1$ has an odd number of divisors (including 1 and itself) if and only if $\sqrt{n}$ is an integer.

Pigeonhole Principle

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

Example

Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Example

How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

Example

Show that for every integer n there is a multiple of n that has only 0s and 1s in its decimal expansion.

Generalized Pigeonhole Principle

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Example

Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

Example

1. How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are selected?
2. How many must be selected from a standard deck of 52 cards to guarantee that at least three hearts are selected?

Tree Diagrams

Example

How many bit strings of length four do not have two consecutive 1s?

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Mixed Problems

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Example

Show that a natural number $n \geq 1$ has an odd number of divisors (including 1 and itself) if and only if $\sqrt{n}$ is an integer.

Binomial Formula

Recall that there are $\frac{n!}{(n-k)!}$ 1-1 functions $f : \{1, 2, ..., k\} \to S$ where S is a set with n elements. This means that there are $\frac{n!}{(n-k)!}$ ordered stings of elements of S of length k. These are called k-permutations of S. Any k-element subset of S has $k!$ different orderings (permutations). Therefore, we have

$$\frac{\frac{n!}{(n-k)!}}{k!}$$

k-element subsets of S. They are called the k-combinations of S. Hence, we have

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

k-combinations of S.

Binomial Formula

Let us consider the product $(x+y)^n$. Each term will be of the form $x^p y^q$ where $p+q=n$. When we calculate each term, we can consider the set of words of length n written with the letters x and y. We can write a n-string with k times xs in $\binom{n}{k}$ many ways. Therefore, when we multiply $(x+y)^n = (x+y)(x+y)\cdots(x+y)$, we have $\binom{n}{k}$ many $x^k y^{n-k}$, for each k. Therefore, we have

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Two Special Cases

Corollary

We put $x = y = 1$. Then we have

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Therefore, a set with n elements has 2^n subsets. This is the third proof we have for this very important result.

Two Special Cases

Corollary

Now put $x = 1$ and $y = -1$. Then we have

$$\sum_{k=0}^{n} \binom{n}{k} (-1)^k = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots (-1)^n \binom{n}{n} = 0.$$

Let S be a set with n elements. Let us call a subset $A \subseteq S$ even if it has an even number of elements. Similarly odd subsets are defined. This corollary says that the number of even subsets of S and the number of odd subsets of S are equal.

Binomial Formula

There are many properties of these coefficients: For example, one can show very easily that

1. $\binom{n}{k} = \binom{n}{n-k}$,
2. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. Hence we have the Pascal triangle..

Theorem (Pascal's Identity)

Let n and k be positive integers with $k \leq n$. Then we have

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof.

□

Binomial Formula

Theorem (Vandermonde's Identity)

m, n, r are positive integers with $r \leq m$ and $r \leq n$. Then we have

$$\binom{m+n}{k} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}.$$

Proof.

□

Corollary

Let n be a positive integer. Then we have

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

Binomial Formula

One can use combinatorial proofs to get more on binomial coefficients. An example is the following:

Theorem

n, r are positive integers with $r \leq n$. Then we have

$$\binom{n+1}{r+1} = \sum_{j=r}^{n} \binom{j}{r}.$$

Binomial Formula

Idea:
$\binom{n+1}{r+1}$ is the number of bit strings of length $n+1$ containing $r+1$ ones.

There is one with the last 1 in the position $r+1$. There are $\binom{r+1}{r}$ with the last one in the position $r+2$, there are $\binom{r+2}{r}$ with the last 1 in the porisitin $r+2$, etc.. The sum of all these should be equal to $\binom{n+1}{r+1}$. □

Selected Exercises

(18) Show that if n is a positive integer, then

$$\binom{n}{0} < \binom{n}{1} < \binom{n}{2} < \cdots < \binom{n}{\lfloor \frac{n}{2} \rfloor} = \binom{n}{\lceil \frac{n}{2} \rceil} > \cdots > \binom{n}{n-1} > \binom{n}{0}.$$

(19) Show that $\binom{n}{k} \leq 2^n$ whenever n is a positive integer and $0 \leq k \leq n$.

(20) a If n is an integer with $n > 1$, show that $\binom{n}{\lceil \frac{n}{2} \rceil} \geq \frac{2^n}{n}$.

b If n is a positive integer, show that $\binom{2n}{n} \geq \frac{4^n}{2n}$.

(25) If $n, k \in \mathbb{N}$ with $1 \leq k \leq n$, we have

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Selected Exercises

(26) Prove that

$$\binom{n}{r}\binom{r}{k} = \binom{n}{k}\binom{n-k}{r-k}$$

whenever $k, n, r \in \mathbb{N}$ with $k \leq r \leq n$.

(31) [Hockeystick Identity] Show that

$$\sum_{k=0}^{r} \binom{n+k}{k} = \binom{n+r+1}{r}$$

whenever n and r are positive integers.

(33) Show that $\displaystyle\sum_{k=1}^{n} k\binom{n}{k} = n \cdot 2^{n-1}$.

Selected Exercises

(37) One can go either 1 unit in the $+x$ direction or 1 unit in the $+y$ direction at each step. Let $A = (0, 0)$, $B = (m, n)$. If $n(A, B)$ denotes the number of paths from A to B, we have

$$n(A, B) = \binom{n+m}{m}.$$

(40) By using ex. 37 show that

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

(41) Use ex. 37 to prove

$$\sum_{k=0}^{r} \binom{n+k}{k} = \binom{n+r+1}{r}.$$

(42) Prove that

$$\sum_{k=0}^{n} k^2 \binom{n}{k} = n(n+1)2^{n-2}$$

if n is a positive integer.

Permutations with Repetitions

Theorem

The number of r-permutations of a set with n elements with repetitions allowed is n^r.

Combinations with Repetitions

Example

We have three types of fruits. We want to choose four fruits. How many possibilities are there? (There are enough of each type)

The problem can be considered as follows: We have three boxes. They are separated by two sticks: $|$ and $|$. Suppose we want to place four fruits $**\,|\,*\,|\,**$ into these boxes.

Combinations with Repetitions

For example, $**|*|*$ represents the situation where we have two fruits of the first type, one of the second type and one of the third type. Similarly, $|**|**$ represent the collection of two fruits of type 2 and two fruits of type 3. Therefore, combinatorially the problem is the same as the following: We have 4+2=6 places. We have to choose two places for the sticks. Therefore we have $\binom{6}{2} = 15$ choices.

Combinations with Repetitions

Theorem

The number of r-combinations of a set with n elements is

$$\binom{n+r-1}{n-1} = \binom{n+r-1}{r}$$

if repetition is allowed.

Example

There are 4 kinds of cookies. 6 cookies are to be chosen.

Now there are four boxes separated by 3 sticks "|" and 6 cookies $*$ to be distributed.

Therefore we have $\binom{9}{3}$ choices.

Combinations with Repetitions

Example

We want to find the number of solutions of

$$x + y + z = 11$$

where $x, y, z \geq 0$.

In this case we have three boxes separated by two sticks. We also have eleven $*$'s. Therefore we have $\binom{13}{2}$ choices.

Example

We consider the same equation with the constraints $x \geq 1, y \geq 2, z \geq 3$. $(\binom{7}{2})$

Combinations with Repetitions

Example

How many strings are there with the letters of SUCCESS?

Theorem

We distribute n objects into k boxes. We have

$$\binom{n}{n_1, n_2, ..., n_k} = \frac{n!}{n_1! \cdot n_2! \cdots n_k!}$$

choices.